

Saurabh singh ☆☆☆☆☆
H-30, Madhuban CHS, Sector 22, koperkhairane, Navi Mumbai,
talk_pks@yahoo.co.in,
02227541862,

Report of Evidence of Use

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Patent Number: USUS9110563B2

Claim Number: (1

What is claimed is: 1. A system for a 3 dimensional (3-D) user interface, the system comprising: one or more 3-D projectors configured to display an image at a first location in a 3-D coordinate system; one or more sensors configured based on quadrilateral angle navigation to sense user interaction with the displayed image and to provide user interaction information; and a digital processor configured: (i) to receive the user interaction information from the one or more sensors, (ii) to correlate the user interaction with the image, and (iii) to provide one or more indications responsive to a correlation of the user interaction with the image, wherein the one or more indications comprise displaying the image at a second location in the 3-D coordinate system.

Element 1:

What is claimed is: 1. A system for a 3 dimensional (3-D) user interface,

Tsao discloses a system for a 3 dimensional (3-D) user interface (For e.g., there is provision for barrier-free interaction with 3D images.).

See e.g., **Abstract**("This invention relates generally to new and improved methods and apparatus for displaying volumetric 3D images with colors and for barrier-free interaction with the 3D images.").

Element 2:

the system comprising: one or more 3-D projectors

Tsao disclosesthe system which comprises one or more 3-D projectors (For e.g., this prior art discloses a volumetric image (3D image) projector.).

*See e.g.,***12:49-55**("By principle of optics, the volumetric images displayed in the display volume can be projected as real images floating in mid air **2350** near the optical axis of the concave reflector, as shown in the figure. The image is 3D and appears popping outside of the reflector. This system is hereby called"volumetric image projector"."). Fig. 23

Element 3:

configured to display an image at a first location in a 3-D coordinate system;

Tsao discloses one or more 3D projectors configured to display an image at a first location in a 3-D coordinate system (For e.g., 3D images are projected which seem floating in mid air 2350 near the optical axis of the concave reflector.).

*See e.g.,***12:49-55**("By principle of optics, the volumetric images displayed in the display volume can be projected as real images floating in mid air **2350** near the optical axis of the concave reflector, as shown in the figure. The image is 3D and appears popping outside of the reflector. This system is hereby called "volumetric image projector"."). Fig.23

Element 4:

one or more sensors configured based on quadrilateral angle navigation

Tsao discloses one or more sensors configured based on quadrilateral angle navigation (For e.g.,sensors such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface, can be integrated with the system or configured based on quadrilateral angle navigation. Position tracking system can be based on quadrilateral navigation.).

*See e.g.,***12:55-58** ("A 3D input device **2360**, such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface, can be integrated with the system to allow the user **2370** interact directly with the 3D images."). Fig.23

Element 5:

to sense user interaction with the displayed image

Tsao discloses sensors to sense user interaction with the displayed image (For e.g., sensors , such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface, can be integrated with the system to allow the user **2370** interact directly with the 3D images which is possible only because of sensors.).

*See e.g.,***12:55-58** ("A 3D input device **2360**, such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface, can be integrated with the system to allow the user **2370** interact directly with the 3D images."). fig. 23

Element 6:

and to provide user interaction information;

Tsao discloses sensors, to provide user interaction information (For e.g., sensors such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface are used to allow the user 2370 interact directly with the 3D images. These sensors are 3D input devices 2360 which enable user to handle data or to manipulate.).

*See e.g.,***12:41-45** ("The volumetric 3D display systems can further be modified to provide a better human systems interface. One desirable feature is to allow the user to "reach into" the 3D data image to perform desired data handling or manipulation."); **12:55-58** ("A 3D input device **2360**, such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface, can be integrated with the system to allow the user **2370** interact directly with the 3D images."). Fig. 23

Element 7:

and a digital processor

Tsao discloses a digital processor (For e.g., the position tracker is connected to a computer.).

*See e.g.,***18:62-65** ("providing a plural position tracker for tracking the relative position and orientation of said physical end with respect to the volumetric 3D display; the position tracker being connected to said computer;").

Element 8:

configured: (i) to receive the user interaction information from the one or more sensors,

Tsao discloses a digital processor configured to receive the user interaction information from the one or more sensors (For e.g., the position tracking system senses the positions and it is an input device 2360 connected to said computer).

*See e.g.,***12:55-58** ("A 3D input device **2360**, such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface, can be integrated with the system to allow the user **2370** interact directly with the 3D images."); **18:62-65** ("providing a plural position tracker for tracking the relative position and orientation of said physical end with respect to the volumetric 3D display; the position tracker being connected to said computer;").

Element 9:

(ii) to correlate the user interaction with the image,

Tsao discloses a digital processor configured to correlate the user interaction with the image (For e.g., input 3D devices 2360, such as a position tracking system allows user to handle the

image data or manipulate it. This input device is connected to a computer to provide data. Since, this system allows direct interaction (data handling/ manipulation) with the 3D image data, this implies that input data to the computer is processed or correlated to reflect desired data handling or manipulation.).

*See e.g.,***12:41-45** ("The volumetric 3D display systems can further be modified to provide a better human systems interface. One desirable feature is to allow the user to "reach into" the 3D data image to perform desired data handling or manipulation."); **12:55-58** ("A 3D input device **2360**, such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface, can be integrated with the system to allow the user **2370** interact directly with the 3D images."); **18:62-65** ("providing a plural position tracker for tracking the relative position and orientation of said physical end with respect to the volumetric 3D display; the position tracker being connected to said computer;").

Element 10:

and (iii) to provide one or more indications

Tsao discloses a digital processor configured toprovide one or more indications (For e.g., activities such as"reach into" the 3D data, data handling, manipulation or direct interaction simply indicate that there are indications of interaction to be felt/ viewed that there areinteractions involved.).

*See e.g.,***12:41-58**("The volumetric 3D display systems can further be modified to provide a better human systems interface. One desirable feature is to allow the user to "reach into" the 3D data image to perform desired data handling or manipulation.FIG. 23 illustrates one preferred embodiment of such a "barrier free" display system, which combines a volumetric display **2310** with a concave mirror **2320**. The display volume of the volumetric display system is placed at a location near the optical axis of a concave reflector **2321**. By principle of optics, the volumetric images displayed in the display volume can be projected as real images floating in mid air **2350** near the optical axis of the concave reflector, as shown in the figure. The image is 3D and appears popping outside of the reflector. This system is hereby called "volumetric image projector". A 3D input device **2360**, such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface, can be integrated with the system to allow the user **2370** interact directly with the 3D images.");

For e.g., in fig. 24, user is able to pick up a tissue surface 2490 with the help of computer programming (processor), and control. Indication of this action is visible in real time as shown in figure. This "virtual manipulator" concept can also be used in the "volumetric image projector" shown in fig. 23.

*See e.g.,***13:36-49** ("Haptic interface such as force-feedback systems can be integrated with this system to create "virtual tools". For example, as illustrated in FIG. 24, the manipulating device is a virtual tweezers with two force-feedback actuation points 2472 and 2473. The user can use two fingers to hold the two action points and applies force. The "virtual end" 2471 can be defined as two tips of the tweezers, which move as the user manipulates the two actuation points. With a suitable computer programming and control, the user can then use the virtual tweezers to "pick up" a tissue surface 2490, with both the tweezers tips and the tissue in the form of volumetric 3D images in the display volume. The "virtual manipulator" concept can, of course, also be used in the "volumetric image projector"."). Fig. 24

Element 11:

responsive to a correlation of the user interaction with the image,

Tsao discloses indications responsive to a correlation of the user interaction with the image (For e.g., user 2370 can directly interact with 3D images displayed using “volumetric image projector”. There are input sensors connected to the computer for data processing. In fig. 24, user is shown picking up a tissue surface 2490, with both the tweezers tips and the tissue in the form of volumetric 3D images in the display volume. Which can also be applied for “volumetric image projector”.).

*See e.g.,***12:49-58** (“By principle of optics, the volumetric images displayed in the display volume can be projected as real images floating in mid air **2350** near the optical axis of the concave reflector, as shown in the figure. The image is 3D and appears popping outside of the reflector. This system is hereby called “volumetric image projector”. A 3D input device **2360**, such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface, can be integrated with the system to allow the user **2370** interact directly with the 3D images.”); **18:62-65** (“providing a plural position tracker for tracking the relative position and orientation of said physical end with respect to the volumetric 3D display; the position tracker being connected to said computer;”).

*See e.g.,***13:36-49** (“Haptic interface such as force-feedback systems can be integrated with this system to create “virtual tools”. For example, as illustrated in FIG. 24, the manipulating device is a virtual tweezers with two force-feedback actuation points 2472 and 2473. The user can use two fingers to hold the two action points and applies force. The “virtual end” 2471 can be defined as two tips of the tweezers, which move as the user manipulates the two actuation points. With a suitable computer programming and control, the user can then use the virtual tweezers to “pick up” a tissue surface 2490, with both the tweezers tips and the tissue in the form of volumetric 3D images in the display volume. The “virtual manipulator” concept can, of course, also be used in the “volumetric image projector”.”).

Element 12:

wherein the one or more indications comprise displaying the image

Tsao discloses the one or more indications comprise displaying image (For e.g., in fig. 23, “volumetric image projector” is used to project the volumetric image. Direct interaction with 3D image is possible. In fig. 24, user executes “picking up” ofa tissue surface 2490, in the display volume which is visible in the fig. 24. According to this prior art,same concept can be applied for “volumetric image projector” i.e. response can be seen in display as an indication of interaction with the 3D image.).

See e.g., **12:49-58**(“By principle of optics, the volumetric images displayed in the display volume can be projected as real images floating in mid air **2350** near the optical axis of the concave reflector, as shown in the figure. The image is 3D and appears popping outside of the reflector. This system is hereby called “volumetric image projector”. A 3D input device **2360**, such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface, can be integrated with the system to allow the user **2370** interact directly with the 3D images.”); **18:62-65** (“providing a plural position tracker for tracking the relative position and orientation of said physical end with respect to the volumetric 3D display; the position tracker being connected to said computer;”).

*See e.g.,***13:36-49** (“Haptic interface such as force-feedback systems can be integrated with this system to create “virtual tools”. For example, as illustrated in FIG. 24, the manipulating device is a virtual tweezers with two force-feedback actuation points 2472 and 2473. The user can use two fingers to hold the two action points and applies force. The “virtual end” 2471 can be

defined as two tips of the tweezers, which move as the user manipulates the two actuation points. With a suitable computer programming and control, the user can then use the virtual tweezers to “pick up” a tissue surface 2490, with both the tweezers tips and the tissue in the form of volumetric 3D images in the display volume. The “virtual manipulator” concept can, of course, also be used in the “volumetric image projector”.”).

Element 13:

at a second location in the 3-D coordinate system.

Tsao discloses indication of interaction with 3D image in the display at a second location in the 3D coordinate system (For e.g., in fig. 23, “volumetric image projector” is used to project the volumetric image. Direct interaction with 3D image is possible. In fig. 24, user executes “picking up” ofa tissue surface 2490, in the display volume which is visible in the fig. 24. According to this prior art, same concept can be applied to “volumetric image projector”. Thus, according to this prior art, picking up the tissue surface causes movement of the tissue to a second location which can be projected via “volumetric image projector”).

See e.g., **12:49-58** (“By principle of optics, the volumetric images displayed in the display volume can be projected as real images floating in mid air **2350** near the optical axis of the concave reflector, as shown in the figure. The image is 3D and appears popping outside of the reflector. This system is hereby called “volumetric image projector”. A 3D input device **2360**, such as a position tracking system, a 3D coordinate measurement arm, or a haptic interface, can be integrated with the system to allow the user **2370** interact directly with the 3D images.”);

*See e.g.,***13:36-49** (“Haptic interface such as force-feedback systems can be integrated with this system to create “virtual tools”. For example, as illustrated in FIG. 24, the manipulating device is a virtual tweezers with two force-feedback actuation points 2472 and 2473. The user can use two fingers to hold the two action points and applies force. The “virtual end” 2471 can be defined as two tips of the tweezers, which move as the user manipulates the two actuation points. With a suitable computer programming and control, the user can then use the virtual tweezers to “pick up” a tissue surface 2490, with both the tweezers tips and the tissue in the form of volumetric 3D images in the display volume. The “virtual manipulator” concept can, of course, also be used in the “volumetric image projector”.”).

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We understand that for the purpose of assisting you in collecting data, you may disclose to us certain confidential information that you want to keep confidential; therefore, for good and valuable consideration, the receipt and sufficiency of which is hereby acknowledged, we agree as follows:

1. For a period of twelve (36) months from the date we receive your confidential information ("Effective Date"), we will not use or disclose the information, except: (a) in the event that such information (i) comes into the public domain not through us; (ii) is already known to us at the time of disclosure; (iii) is lawfully otherwise learnt or independently developed by us; or (iv) is required to be disclosed by law.

2. Since a breach of our obligations hereunder may result in irreparable harm to you, we agree that upon such breach or threat thereof, you shall be entitled to specific performance and other injunctive relief without posting a bond, and in the event of disclosures required by court order, we shall promptly notify you of the proceeding.

3. This agreement supersedes all prior discussions and writings and constitutes the entire agreement between you and us with respect to the subject matter hereof. This Agreement shall be binding upon the successors and assigns of the respective parties, and may not be modified except by an instrument executed by both parties.

Saurabh singh